AI-based intent-aware garbage intrusion detection and alert system for swimming pools

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Project Proposal Report

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1. DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Prof. Anuradha Karunasena	13/03/2026	

2. ABSTRACT

Ensuring hygienic and safe conditions in public and private swimming pools is a major worry for facility managers as well as users. Garbage can significantly damage swimming pools in a number of ways, including affecting water quality, increasing private pool maintenance costs and even harming the health of swimmers. Manual monitoring tasks are usually performed by lifeguards or surveilling personnel in most swimming pools, which can be ineffective and error-prone. Even though there are still some of modern monitoring systems that deploy computer vision methods to identify objects in supervised areas, usually these techniques provide only object detection without any consideration for the context or human activity related to the event.

As a result, none of the existing object detection systems are capable of separating an intentional garbage discarding from an unintentional drop of objects, resulting in

to false alarms and ineffective monitoring processes. Moreover, prevalent pool monitoring systems mainly focus either on water quality or drowning detection with limited research done in garbage intrusion detection.

Specifically, to overcome these limitations, this research proposes an artificial intelligence (AI)-based intent-aware garbage intrusion detection and alerting system that will automatically identify the waste objects entering a swimming pool along with analyzing the human actions corresponding to the intrusion event. We are proposing a system that would be able to record real-time video streams

3. ACKNOWLEDGEMENT

I want to sincerely thank my research supervisor for their unwavering support, insightful advice, and encouragement during the writing of this research proposal. Their assistance and professional guidance made it much easier for me to comprehend the research procedure and raise the caliber of this work.

Additionally, I would like to express my gratitude to the Sri Lanka Institute of Information Technology's Department of Information Technology for providing the resources and academic support needed to complete this research proposal.In addition, I would like to express my sincere gratitude to all of the professors and academic staff who have helped me along the way and offered insightful information that helped me formulate this research idea.

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7. LIST OF ABBREVIATIONS

AI - Artificial Intelligence

ML - Machine Learning

CNN - Convolutional Neural Network

YOLO - You Only Look Once

YOLOv8 - You Only Look Once version 8

LSTM - Long Short-Term Memory

IoT - Internet of Things

API - Application Programming Interface

CCTV - Closed-Circuit Television

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1. INTRODUCTION

1.1 Background & Context

Recreational centers, hotels, sports complexes, and residential buildings all frequently have swimming pools. To protect swimmers' health and safety, swimming pools must be kept clean and hygienic. Garbage items like plastic bottles, food wrappers, and other waste materials getting into the swimming pool is one of the frequent issues that pool managers deal with. These intrusions have the potential to lower water quality, raise maintenance expenses, and make pool users' surroundings unpleasant.to detect undesired activities or objects entering the pool area, lifeguards or security cameras manually monitor swimming pools. However, because lifeguards might not always keep an eye on everything going on around the pool, manual monitoring is not always dependable.
In order to identify objects in surveillance environments, a number of contemporary monitoring systems have incorporated computer vision and artificial intelligence techniques. These systems use object detection frameworks like YOLO and deep learning models like convolutional neural networks (CNN) to identify objects within a scene. Nevertheless, a lot of these systems only concentrate on object detection; they don't examine human behavior or the motivation behind the incident. As a result, they might sound an alarm even if something unintentionally falls into the pool, creating needless alarms.
This study suggests an AI-based intent-aware garbage intrusion detection and alert system for swimming pools as a solution to this problem. The planned system will use security cameras positioned throughout the pool area to record live video streams. The system will identify trash items entering the pool using computer vision techniques and assess human behavior to ascertain whether the trash intrusion was deliberate or unintentional. The system seeks to increase monitoring effectiveness, decrease false alerts, and improve swimming pool safety and hygiene by fusing object detection and human action recognition.

1.2 Literature Review

Intelligent monitoring systems that can analyze video streams in real time have been made possible by recent advancements in computer vision and artificial intelligence. Numerous studies have investigated the use of deep learning techniques for object detection, human action recognition, and environmental monitoring.In surveillance settings, object detection has grown in popularity as a method of object identification. The You Only Look Once (YOLO) framework, developed by Joseph Redmon's research, enables highly accurate real-time object detection. Smart surveillance systems have made extensive use of YOLO-based models to identify objects like trash, cars, and suspicious objects. In a similar vein, research on accurate object recognition tasks in complex environments has made extensive use of the Faster R-CNN model.

To comprehend human behavior in surveillance footage, numerous researchers have investigated human action recognition in addition to object detection. Research utilizing recurrent and convolutional neural networks has demonstrated encouraging outcomes in recognizing actions like running, walking, and object throwing. Security monitoring

and activity analysis systems have employed methods like CNN-LSTM architectures and pose-based recognition models.

For environmental monitoring, a number of researchers have also created intelligent waste detection systems. These systems identify trash in public areas like streets, parks, and beaches using deep learning models. These systems can identify objects like plastic bottles, food wrappers, and containers by using computer vision models trained on datasets containing waste objects.

Apart from research-based approaches, some existing products and smart surveillance solutions have been developed for monitoring environments. AI-based surveillance platforms such as OpenCV and deep learning frameworks like TensorFlow and PyTorch have been widely used to develop intelligent monitoring applications. These tools provide the capability to build object detection models and real-time video analysis systems.Despite the progress in AI-based surveillance technologies, several limitations still exist in current systems. Many monitoring solutions rely only on object detection and cannot tell the difference between intentional and accidental events. As a result, these systems often trigger false alerts when objects are dropped by accident. Furthermore, many current pool monitoring systems mainly focus on water quality checks or drowning detection instead of identifying garbage intrusion events.

1.3 Research Gap

Although several studies have looked into object detection and recognizing human actions using artificial intelligence, a significant gap remains in applying these technologies for intent-aware garbage intrusion detection in swimming pool areas. Most current monitoring systems depend on manual supervision or traditional surveillance cameras, which are not effective for continuous monitoring of swimming pools. Some research has applied computer vision techniques to detect garbage or foreign objects, but these systems usually only identify the presence of objects without considering the human behaviour associated with those events. As a result, existing systems cannot tell the difference between someone intentionally throwing away trash and someone accidentally dropping an object. This leads to many false alerts and ineffective monitoring. Additionally, many pool monitoring technologies mainly focus on water quality or drowning detection, while garbage intrusion detection has not been widely researched.

Thus, there is a need for a smart monitoring solution that can analyse both the detected object and the human action causing the intrusion. Combining object detection with human action recognition can provide a clearer understanding of intrusion events and help the system prioritize high-risk situations.

To fill this research gap, this study suggests an AI-based intent-aware garbage intrusion detection and alert system for swimming pools. The system will merge

computer vision garbage detection with human action analysis to spot intentional garbage intrusion events and send real-time alerts to pool management. By incorporating intent awareness, the proposed system aims to reduce false alarms, improve monitoring efficiency, and help maintain cleaner and safer swimming pool environments.

2. OBJECTIVE

B. 2.1 Main Objective

The main goal of this research is to create an AI-based system that can detect garbage entering swimming pools and alert management. This system will automatically monitor pool environments, identify garbage objects, and assess human actions linked to these incidents. By using computer vision for object detection along with techniques for recognising human activities, the system will determine if the intrusion is intentional or accidental. It will also send real-time alerts to pool managers when it detects suspicious or high-risk garbage intrusion events. Ultimately, this research aims to improve pool hygiene, boost monitoring efficiency, and lessen the drawbacks of traditional manual supervision methods.

C. 2.2 Specific Objectives

First, the research aims to capture and preprocess real-time video streams from surveillance cameras around the swimming pool area. This step ensures that the collected video data is suitable for further analysis using computer vision techniques.

Second, the study plans to develop a computer vision-based object detection model that can identify garbage items like plastic bottles, food wrappers, and other waste that may enter the swimming pool.

Third, the research focuses on implementing a human action recognition module. This module will analyze video sequences to identify actions performed by individuals near the pool, especially those related to throwing or dropping objects.

Fourth, the study aims to combine object detection with human action analysis to determine the intent behind garbage intrusion events. This will help distinguish between intentional garbage disposal and accidental drops.

Fifth, the proposed system will send real-time alerts and notifications to inform lifeguards or pool management when high-risk garbage intrusion events occur.

Finally, the system will keep a record of intrusion events and alert history. This information can be used for monitoring, analysis, and improving the system's performance.

3. METHODOLOGY

This section explains the research method, technical methods, tools and platforms, data collection process, and evaluation techniques used to create the AI-based intent-aware garbage intrusion detection and alert system for swimming pools.

3.1 Research Approach

This research uses a Design Science Research (DSR) approach. This method focuses on creating and developing a new technological solution to address a real-world issue. The study aims to design an intelligent monitoring system that can detect garbage intrusion events in swimming pool areas while examining the human actions linked to these events. The system will be developed, implemented, and tested to assess its effectiveness in spotting garbage objects and recognizing intentional intrusion behavior. The design science approach fits this research because it encourages the creation and assessment of new technological tools like AI-based monitoring systems.

3.2 Technical Methods and Algorithms

The proposed system will use computer vision and deep learning techniques to analyze video streams from surveillance cameras. It will include multiple parts, such as video preprocessing, object detection, human action recognition, and intent analysis.

First, the system will capture real-time video streams and preprocess them with image processing techniques to extract frames and eliminate noise. After preprocessing, an object detection model will recognize garbage items like plastic bottles, food wrappers, and other waste entering the swimming pool area. We will consider deep learning models like YOLO (You Only Look Once) for their high accuracy and real-time detection capabilities.

Besides object detection, the system will have a human action recognition module to examine the behavior of people near the pool. We will use deep learning methods like Convolutional Neural Networks (CNN) combined with sequence analysis models to identify actions like throwing or dropping objects.

Finally, an intent analysis mechanism will combine the results of object detection and human action recognition to decide if the garbage intrusion event was intentional or accidental. When the system detects an intentional intrusion, it will send real-time alerts to notify pool management.

3.3 Tools and Platforms

Several tools and software platforms will be used to develop and implement the proposed system. The main programming language for this research is Python, as it offers strong support for artificial intelligence and computer vision development.

Computer Vision & Deep Learning

- YOLOv8/YOLOv9 – best choice for real-time object detection(garbage items like bottles, wrappers)
- OpenCV – video stream preprocessing, frame extraction, image augmentation
- MediaPipe or PoseNet – human pose/skeleton tracking to analyze throwing actopns
- TensorFlow/PyTorch – model training and fine-tuning

Human Action Recognition

- SlowFast Networks or I3D – temporal video understanding for action classification
- LSTM or Trasformer-based models – to capture the sequence of human movement over time

Backend & Integration

- Python(FastAPI or Flask) – system backend
- MQTT/WebSocket – real-time alert streaming to the dashboard
- Firebase or Twilio – push notification / SMS alerts to lifegurads
- PostgreSQL/MongoDB – storing intrusion logs and alerts history

Hardware

- IP surveillance cameras(1080p+) – pool perimeter coverage

3.4 Data Collection and Ethical Considerations

The system will need video data for training and testing the object detection and action recognition models. Data will be collected from publicly available datasets related to

waste detection and human activity recognition. Additionally, simulated video data showing garbage intrusion events in swimming pool settings may be used for system testing.

During the research process, ethical issues will be addressed carefully. The system will focus only on analyzing actions related to garbage intrusion events and will not store or misuse personal identity information of individuals captured in video recordings. All collected data will be used only for research purposes and will follow relevant privacy and ethical research guidelines.

3.5 Validation Strategy and Performance Metrics

To evaluate the performance of the proposed system, various validation techniques and performance metrics will be used. The accuracy of the object detection model will be measured with standard evaluation metrics such as precision, recall, and F1-score. These metrics will help assess how effectively the system identifies garbage objects in video frames.

The performance of the human action recognition module will also be evaluated based on its ability to correctly detect throwing or dropping actions. The overall effectiveness of the system will be assessed by measuring the false alert rate, detection speed, and the system's ability to accurately classify intentional intrusion events.

Through these evaluation methods, the research will examine the reliability and efficiency of the proposed AI-based monitoring system in real-world swimming pool environments.

4. HIGH-LEVEL SYSTEM ARCHITECTURE



Figure 1-System Architecture

5. USER REQUIREMENTS

5.1 Stakeholder Identification

- Swimming Pool Management
    - Responsible for keeping the pool clean and safe.
    - Use the system to track garbage intrusion events and receive alerts.

- Lifeguards / Pool Supervisors
    - Observe activities around the pool area.
    - Get immediate alerts when garbage intrusion happens.

- Maintenance Staff
    - Check intrusion logs to find garbage incidents.
    - Carry out cleaning and maintenance tasks as needed.

- Pool Users / Visitors
    - Indirect stakeholders who benefit from a cleaner and safer swimming environment.

5.2 Functional Requirements

1. The system will capture real-time video streams from cameras set up around the swimming pool.
2. The system will preprocess video frames for analysis using computer vision techniques.
3. The system will detect garbage items like plastic bottles, wrappers, and food waste using an AI-based object detection model.
4. The system will analyse human actions around the swimming pool area.
5. The system will identify behaviours like throwing or dropping objects into the pool.
6. The system will combine object detection and action analysis to figure out if the garbage intrusion is intentional or accidental.
7. The system will generate real-time alerts to lifeguards or pool management when a garbage intrusion event occurs.
8. The system will store intrusion event data in a database for monitoring and future analysis.
9. The system will provide a dashboard interface to view alerts and intrusion history.

5.3 Non-Functional Requirements

Non-functional requirements define the quality attributes and performance characteristics expected from the system.

Security:

- The system should ensure secure storage of video data and intrusion records.
- Access to the system should be limited to authorized users.

Performance:

- The system should detect garbage intrusion events almost in real-time.

Reliability:

- The system should work continuously with precise detection performance.

Usability:

- The system interface should be straightforward and easy to use for pool management and lifeguards.

Scalability:

- The system should handle multiple cameras and larger swimming pool facilities if needed.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The proposed AI-Based Intent-Aware Garbage Intrusion Detection and Alert System primarily targets organizations that manage swimming pools. This includes hotels, resorts, sports complexes, public swimming pools, aquatic training centers, and residential buildings with shared pool facilities.

The main customer persona consists of swimming pool managers, hotel management teams, sports facility admins, and maintenance supervisors. They are responsible for keeping pools clean, safe, and running efficiently. These stakeholders need dependable monitoring solutions for maintaining hygienic pool conditions and reducing the need for constant supervision.

6.2 Value Proposition

The proposed system provides a smart, automated solution for detecting garbage in swimming pools. It combines computer vision-based object detection with human action analysis to identify waste entering the pool and assess whether the action was intentional or accidental.

This approach cuts down on false alerts and allows for quick responses from lifeguards or pool management. As a result, the system boosts swimming pool hygiene, improves user safety, and lightens the load of manual monitoring. It also keeps intrusion logs to aid in monitoring and maintenance planning.

6.3 Revenue Model

The system can be offered through several revenue models. One option is software licensing, where swimming pool facilities buy the system for installation and use. Another is a subscription service, allowing customers to pay a monthly or annual fee for access to monitoring, real-time alerts, and system updates. Additional income can come from installation services, maintenance support, and customized system setups for larger facilities.

6.4 Cost Estimation

Implementing this system will involve several costs, including hardware, software development, and system maintenance. Key costs include surveillance cameras for monitoring the pool area, computing hardware or servers for AI processing, cloud storage or database services, and software development expenditures. Operational costs, such as system maintenance, updates, and technical support, must also be included.

6.5 Pricing Strategy

The pricing strategy will aim to make the system affordable for various swimming pool facilities. A tiered pricing model could be introduced, featuring a basic package for small facilities, a standard package with AI detection and alert capabilities, and a premium package with advanced analytics and monitoring dashboards. Pricing may also be available as monthly or yearly subscription plans to lower the initial investment for customers.

6.6 Competitive Advantage

The proposed system has several benefits over traditional pool monitoring methods. Unlike standard surveillance systems that depend on manual monitoring, this solution uses artificial intelligence to automatically detect garbage and analyze human actions. The combination of object detection with intent analysis reduces false alerts and enhances detection accuracy. Additionally, the system provides real-time alerts and keeps logs of intrusions, allowing for better pool management and improved hygiene control.

6.7 Intellectual Property Considerations

The proposed system involves creating original software and AI detection models. Therefore, protecting intellectual property is essential for safeguarding this innovation. Copyright can protect the software, while the unique system architecture and intent-aware detection approach may be eligible for patent protection. Proper management of intellectual property will help maintain the originality of the research and support future commercialization opportunities.

7. BUDGET AND JUSTIFICATION

For this research component, the main goal is not to sell the individual modules separately. Instead, the final result of the research will be a complete AI-based swimming pool monitoring application. The budget set aside for this research primarily supports system development, data processing, testing activities, and the documentation needed to finish the project successfully.

The following table shows the estimated budget needed for developing and evaluating the AI-Based Intent-Aware Garbage Intrusion Detection and Alert System for Swimming Pools.

Table 1- Budget Table

Description	Cost
Electricity	4000.00
Internet usage	5000.00
Communication expenses	2000.00
Transportation	3000.00
Printing and documentation	3000.00
Cloud hosting/backend server for AI system	15000.00
Software tools and development resources	12000.00
Dataset preparation and model training	8000.00
Total	52000.00

7.1 **Budget Justification**

The budget mainly supports the development and testing of the AI-based monitoring system. Expenses for electricity and internet are necessary for system development, AI model training, and ongoing research activities. Costs for communication and transportation help with collaboration among supervisors and possible field visits for system testing. Printing and documentation expenses are needed for preparing research reports and project documentation.

Cloud hosting and backend server costs are essential for system deployment, data storage, and real-time alert services. Software tools and development resources are required for creating the AI models and integrating system components. Additionally, resources for dataset preparation and training are needed to build and improve the object detection and human action recognition models used in the system.

Overall, these expenses ensure the successful development, evaluation, and demonstration of the proposed intelligent garbage intrusion detection system

8. WORK BREAKDOWN STRUCTURE (WBS)

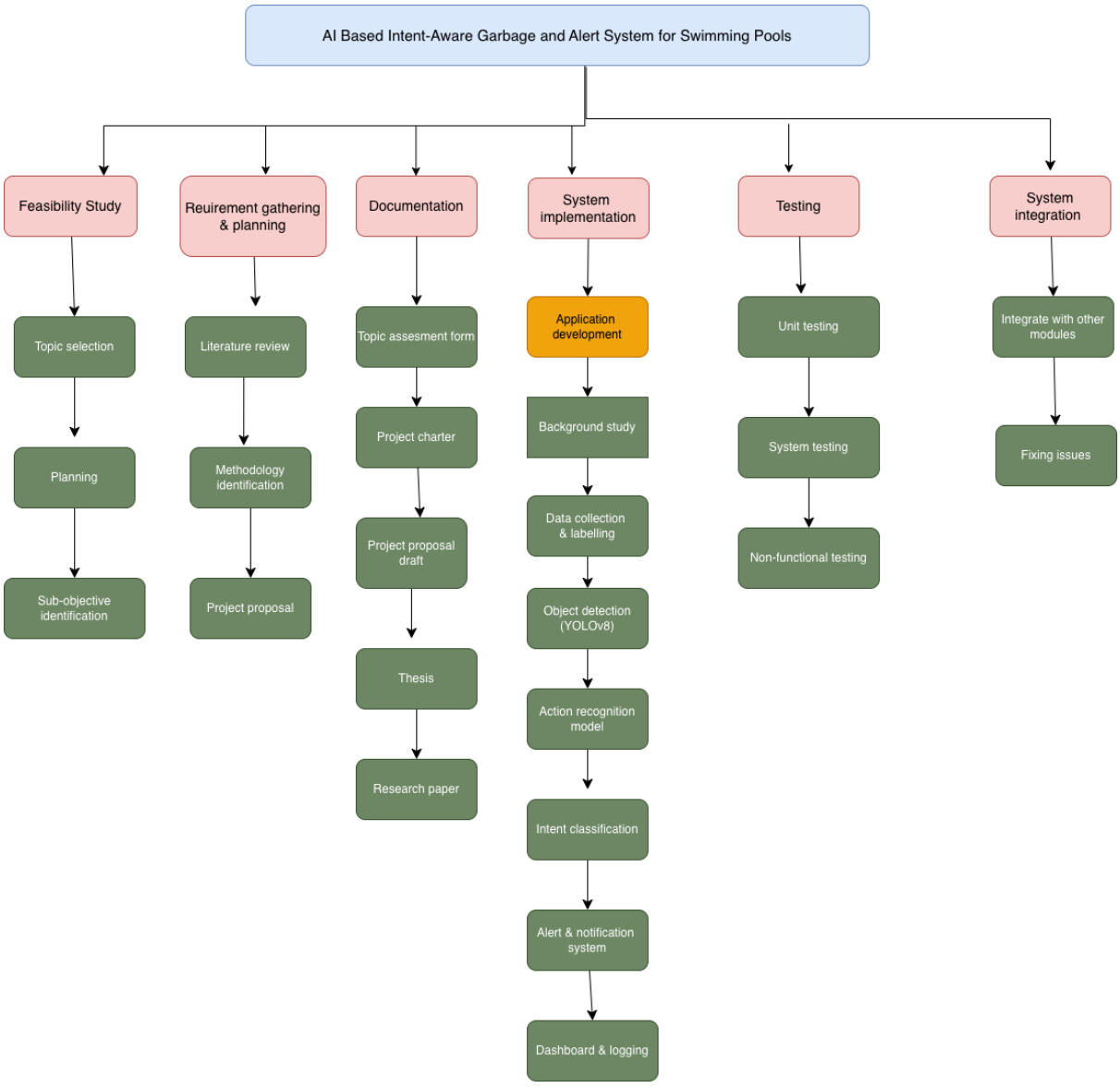


Figure 2 - Work Breakdown Chart

9. GANTT CHART

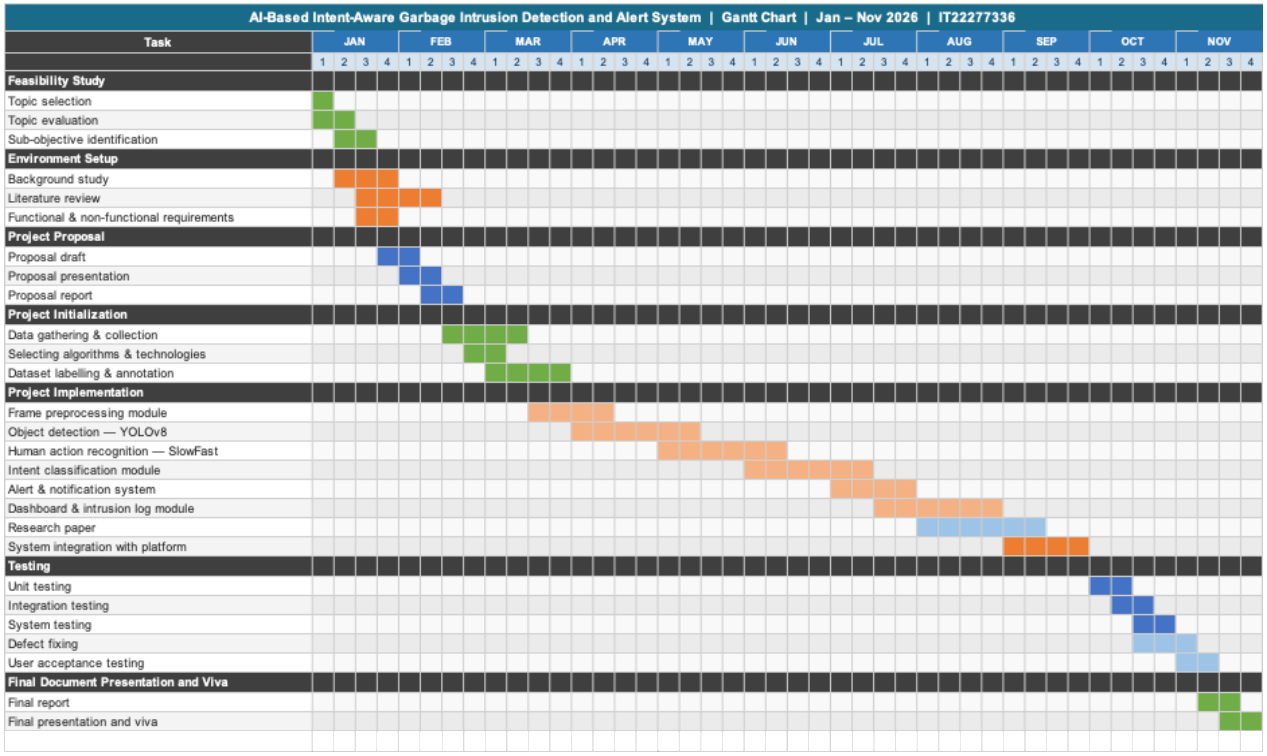


Figure 3 - Gantt Chart

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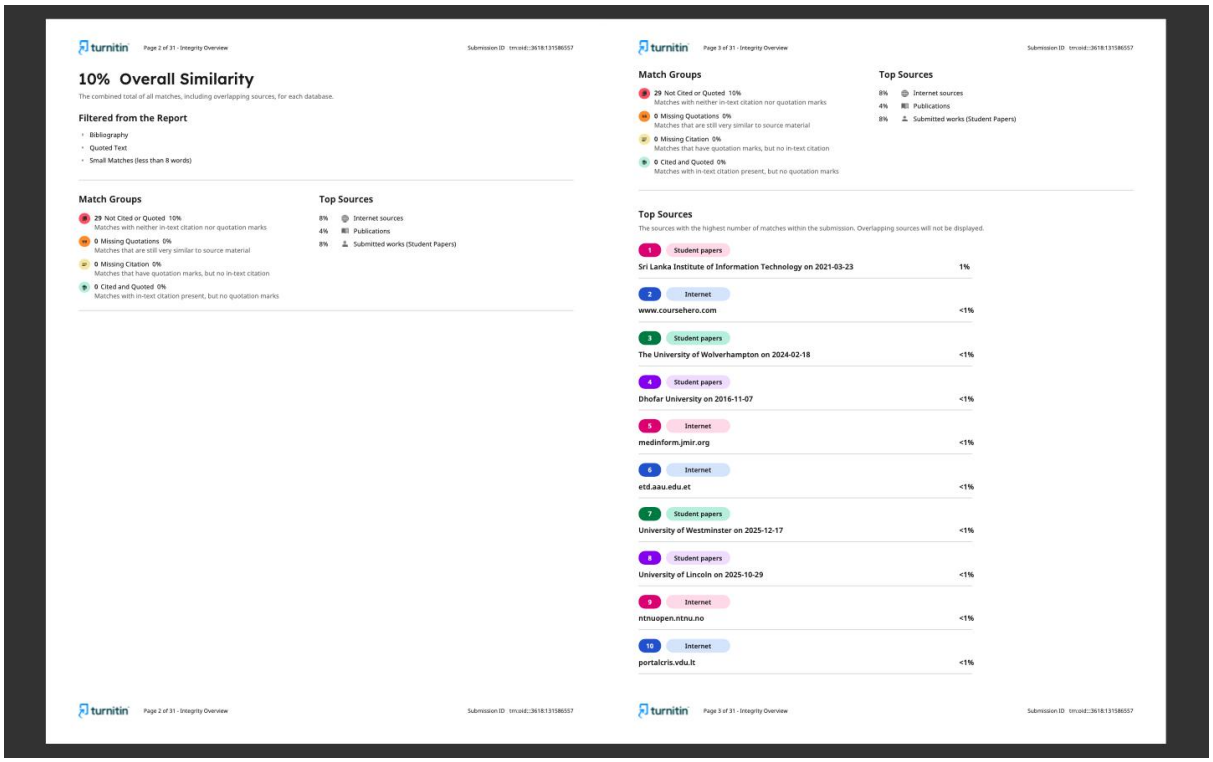
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11.APPENDICES

Appendix A 1 - Similarity Report



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